ormulas & Functions

Practice Instructio

How to use this guide: Download and open your **Excel_E4_Starter** file. Work through each skill using the instructions below. Each skill shows what to do and what the result looks like. Check off each skill on the IS101 reference page.

Skills in This Guide

1. SUM Function
2. AVERAGE Function
3. MIN and MAX Functions
4. COUNT and COUNTA Functions
5. IF Function
6. Nested IF Function
7. COUNTIF Function
8. SUMIF Function
9. LEFT, RIGHT, MID Functions
10. CONCAT and TEXTJOIN Functions
11. TODAY and NOW Functions
12. YEAR, MONTH, DAY Functions
13. VLOOKUP Function
14. XLOOKUP Function
15. IFERROR Function

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1 SUM Function

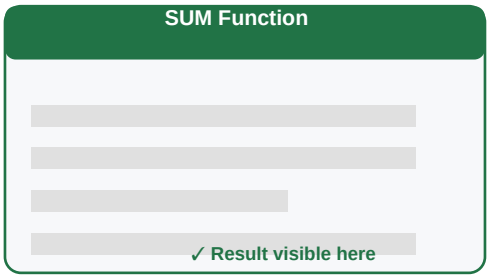
Formulas

Your Task: Calculate total revenue by summing all values in the Amount column.

STEP-BY-STEP INSTRUCTIONS

1. Click the result cell (J35 or a summary cell).
2. Type =SUM(then select the Amount range →) → Enter.
3. Or press Alt+= to AutoSum the column above.

WHAT THE RESULT LOOKS LIKE



The cell displays the total. The formula bar shows =SUM(range).

2 AVERAGE Function

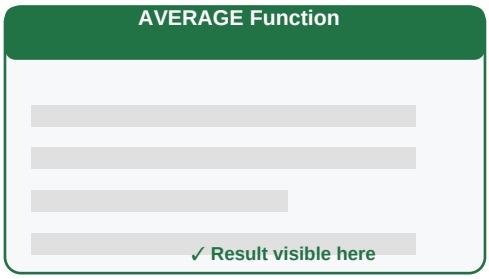
Formulas

Your Task: Calculate the average order value across all transactions.

STEP-BY-STEP INSTRUCTIONS

1. Click the result cell.
2. Type =AVERAGE(→ select the Amount range →) → Enter.

WHAT THE RESULT LOOKS LIKE



The cell shows the arithmetic mean of the range. Formula bar shows =AVERAGE(range).

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3 MIN and MAX Functions

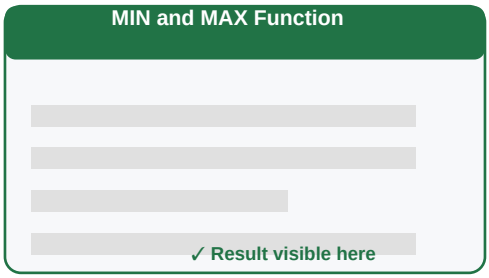
Formulas

Your Task: Find the smallest and largest order amounts in the dataset.

STEP-BY-STEP INSTRUCTIONS

1. For MIN: click result cell → type =MIN(→ select range →) → Enter.
2. For MAX: click result cell → type =MAX(→ select range →) → Enter.

WHAT THE RESULT LOOKS LIKE



One cell shows the smallest order; another shows the largest. Both update when data changes.

4 COUNT and COUNTA Functions

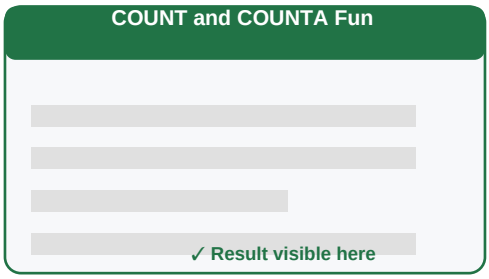
Formulas

Your Task: Count numeric order entries with COUNT and non-empty customer names with COUNTA.

STEP-BY-STEP INSTRUCTIONS

1. For COUNT: =COUNT(→ select number range →) → Enter.
2. For COUNTA: =COUNTA(→ select text range →) → Enter.

WHAT THE RESULT LOOKS LIKE



COUNT returns the number of numeric cells. COUNTA returns the number of all non-empty cells.

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5 IF Function

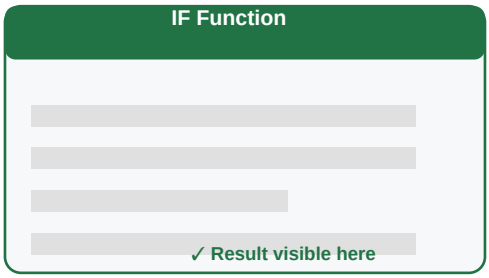
Formulas

Your Task: Flag orders over $500 as "High Value" and all others as "Standard" in the Grade column.

STEP-BY-STEP INSTRUCTIONS

1. Click the Grade cell (H3).
2. Type: =IF(E3>500,"High Value","Standard") → Enter.
3. Copy the formula down to apply to all rows.

WHAT THE RESULT LOOKS LIKE



Each cell shows "High Value" or "Standard" based on the order amount in column E.

6 Nested IF Function

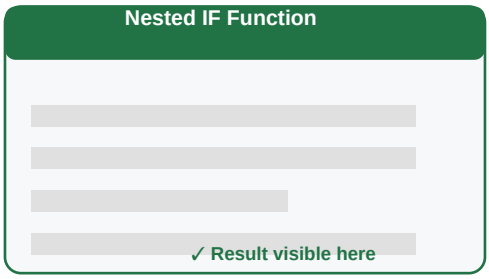
Formulas

Your Task: Assign A, B, or C grades: A for orders over $1000, B for over $500, C for all others.

STEP-BY-STEP INSTRUCTIONS

1. Click the Grade cell.
2. Type: =IF(E3>1000,"A",IF(E3>500,"B","C")) → Enter.
3. Copy down to all rows.

WHAT THE RESULT LOOKS LIKE



Each cell shows A, B, or C. The second IF is nested inside the first IF's false argument.

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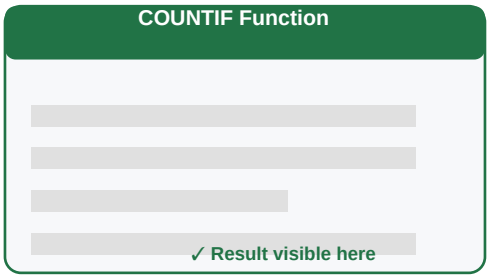
7 COUNTIF Function

Your Task: Count how many orders came from the "Wholesale" category.

STEP-BY-STEP INSTRUCTIONS

1. Click the result cell.
2. Type: =COUNTIF(C3:C32,"Wholesale") → Enter.
3. C3:C32 is the range; "Wholesale" is the criteria.

WHAT THE RESULT LOOKS LIKE



The cell shows the count of orders where the Category column equals "Wholesale".

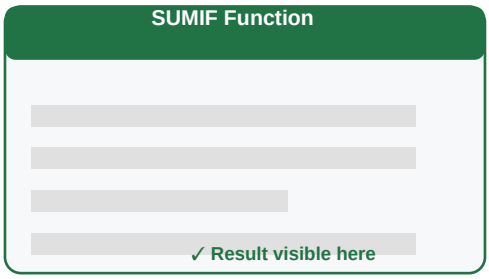
8 SUMIF Function

Your Task: Calculate total revenue from "Corporate" category orders only.

STEP-BY-STEP INSTRUCTIONS

1. Click the result cell.
2. Type: =SUMIF(C3:C32,"Corporate",E3:E32) → Enter.
3. C3:C32 = range to check, "Corporate" = criteria, E3:E32 = range to sum.

WHAT THE RESULT LOOKS LIKE



The cell shows the total of all E column values where C column says "Corporate".

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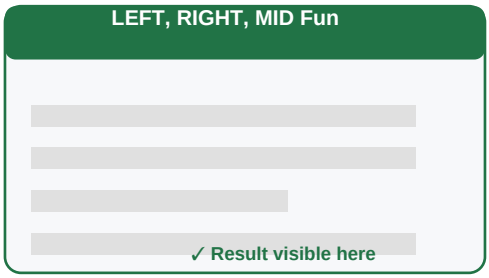
9 LEFT, RIGHT, MID Functions

Your Task: Extract category codes from product codes using LEFT, RIGHT, and MID functions.

STEP-BY-STEP INSTRUCTIONS

1. Click an empty cell in the Text Functions sheet.
2. For LEFT: =LEFT(D2,2) → Enter → copy down.
3. For RIGHT: =RIGHT(D2,3) → Enter.
4. For MID: =MID(D2,4,2) → Enter.

WHAT THE RESULT LOOKS LIKE



Cells extract the specified characters from each product code.

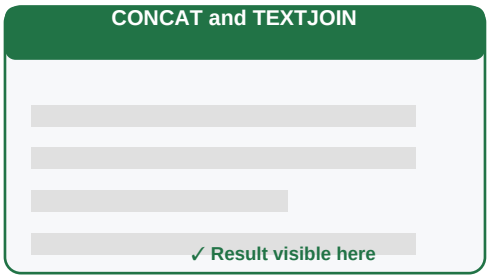
10 CONCAT and TEXTJOIN Functions

Your Task: Combine the First Name and Last Name columns into a Full Name column.

STEP-BY-STEP INSTRUCTIONS

1. Click the Full Name cell (C2 in Text Functions sheet).
2. Type: =CONCAT(A2," ",B2) → Enter.
3. Or use: =TEXTJOIN(" ",TRUE,A2,B2)
4. Copy down to all rows.

WHAT THE RESULT LOOKS LIKE



The Full Name cell shows "FirstName LastName" with a space between.

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11 TODAY and NOW Functions

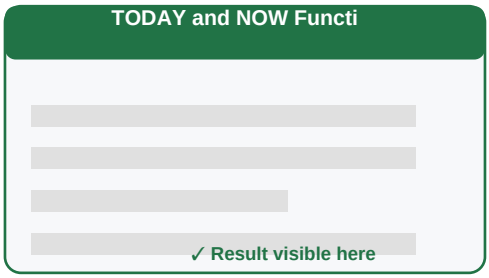
Formulas

Your Task: Display today's date and the current date/time in summary cells.

STEP-BY-STEP INSTRUCTIONS

1. Click the date cell → type =TODAY() → Enter.
2. Click the timestamp cell → type =NOW() → Enter.

WHAT THE RESULT LOOKS LIKE



The date cell shows today's date. The timestamp shows date and time. Both update when the file is opened.

12 YEAR, MONTH, DAY Functions

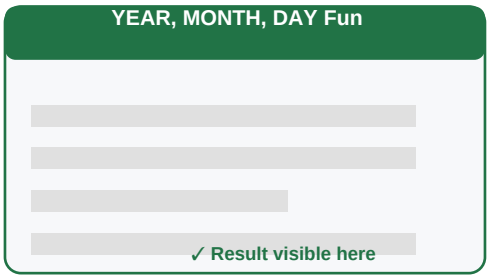
Formulas

Your Task: Extract the year from each order date to populate the Year column.

STEP-BY-STEP INSTRUCTIONS

1. Click the Year cell (G3 in Sales Data sheet).
2. Type: =YEAR(F3) → Enter.
3. Copy down to all rows.
4. Use =MONTH(F3) and =DAY(F3) for month and day.

WHAT THE RESULT LOOKS LIKE



The Year column shows only the 4-digit year extracted from each date.

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13 VLOOKUP Function

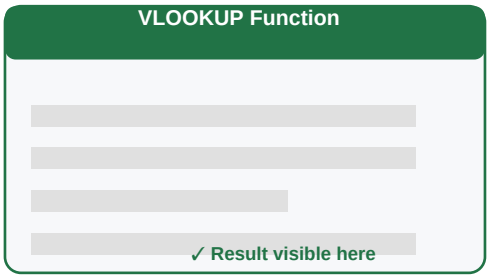
Formulas

Your Task: Look up the unit price from the Price Table using the product code.

STEP-BY-STEP INSTRUCTIONS

1. Click the Unit Price cell (B2 in Lookup Tasks sheet).
2. Type: =VLOOKUP(A2,'Price Table'!A:C,3,FALSE) → Enter.
3. Copy down to all rows.

WHAT THE RESULT LOOKS LIKE



Each cell shows the price for the matching product code. #N/A appears for codes not in the table.

14 XLOOKUP Function

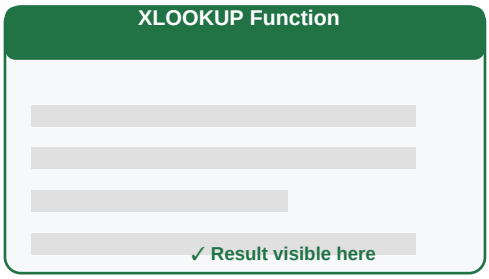
Formulas

Your Task: Find the supplier name for each product code using XLOOKUP.

STEP-BY-STEP INSTRUCTIONS

1. Click the Supplier Name cell (D2 in Lookup Tasks sheet).
2. Type: =XLOOKUP(A2,'Price Table'!A:A,'Price Table'!D:D) → Enter.
3. This finds the Supplier ID, then use another XLOOKUP to get the supplier name.
4. Copy down to all rows.

WHAT THE RESULT LOOKS LIKE



Each cell shows the supplier ID for the matching product code. XLOOKUP is simpler than VLOOKUP.

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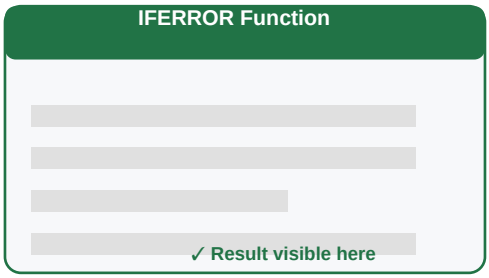
15 IFERROR Function

Your Task: Wrap the VLOOKUP in IFERROR so unmatched codes show "Not Found" instead of #N/A.

STEP-BY-STEP INSTRUCTIONS

1. Click the IFERROR cell (C2 in Lookup Tasks sheet).
2. Type: =IFERROR(VLOOKUP(A2,'Price Table'!A:C,3,FALSE),"Not Found") → Enter.
3. Copy down to all rows.

WHAT THE RESULT LOOKS LIKE



Rows where the lookup succeeds show the price. Unmatched codes show "Not Found" instead of #N/A.

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